When Does Uplift Targeting Pay Off? A Pre-registered, Budget-Constrained Study of the Hillstrom E-mail Experiment

Research Project Report

Simion Polivencu, polivencus@gmail.com

Abstract: Can uplift models choose who gets which promotional e-mail better than simple rules? On the Hillstrom experiment (64,000 customers, three arms), five learners fail to rank customers better than random on spend, and the targeted policy never beats sending the better e-mail to a random subset. A positive control on the same covariates shows that the pipeline recovers a known effect (correlation up to 0.83; +\$0.50 [0.27, 0.75] per customer over random targeting at a 30% budget), and a heterogeneity sweep shows why the real result is a weak null: at Hillstrom's size a single experiment detects a real gain only 5 to 25% of the time unless most of the effect is concentrated in a third of customers. Pre-registered extensions on profit, cross-fitting, policy learning, chance constraints and a budgeted bandit are reported with the same discipline.

1 Introduction

Marketing budgets are finite, so a firm that can send two different promotional e-mails must decide *who* receives *which* e-mail, and how much to spend in total. Uplift modelling (Gutierrez & Gérardy, 2017; Radcliffe, 2007) answers the first question by estimating, for each customer, the conditional average treatment effect (CATE) of each promotion and targeting those with the largest predicted lift. Whether such estimates are *good enough to change a decision* is a different question, and it is the one a business cares about.

Answering it requires connecting four steps that are usually studied separately: a randomised experiment that identifies causal effects; a model of how those effects vary across customers; an optimisation that turns noisy estimates into a feasible allocation under a budget; and an unbiased, uncertainty-quantified evaluation of the resulting policy. Errors compound along this chain: a learner that is "better than random" on a ranking metric may still lose money once the budget is spent, and an optimiser fed noisy estimates systematically over-promises (Smith & Winkler, 2006).

Background. Uplift modelling originated in direct marketing, where Radcliffe (2007) introduced the Qini curve; Gutierrez and Gérardy (2017) connect it to the potential-outcomes framework (Imbens & Rubin, 2015; Rubin, 1974). Multi-treatment uplift is treated by Zhao, Fang, and Simchi-Levi (2017) and benchmarked by Olaya, Coussement, and Verbeke (2020). Meta-learners reduce CATE estimation to supervised learning (Kennedy, 2023; Künzel, Sekhon, Bickel, & Yu, 2019; Nie & Wager, 2021), causal forests adapt random forests to effect estimation (Athey, Tibshirani, & Wager, 2019; Wager & Athey, 2018), and double/debiased machine learning (Chernozhukov et al., 2018) provides the cross-fitting framework used for evaluation. Inverse-propensity weighting (Horvitz & Thompson, 1952) and doubly robust estimators (Dudík, Langford, & Li, 2011; Robins, Rotnitzky, & Zhao, 1994) evaluate a policy from randomised data, and policy learning maximises such estimates directly (Athey & Wager, 2021; Kitagawa & Tetenov, 2018). Assigning one of several costly treatments under a budget is a multiple-choice knapsack (Kellerer, Pferschy, & Pisinger, 2004; Sinha & Zoltners, 1979) with fast Lagrangian solutions (Everett, 1963); chance-constrained programming (Charnes & Cooper, 1959) handles random costs; Thompson sampling (Eckles & Kaptein, 2014; Russo, Van Roy, Kazerouni, Osband, & Wen, 2018; Thompson, 1933) and bandits with knapsacks (Badanidiyuru, Kleinberg, & Slivkins, 2018) address sequential budgeted decisions.

What is new. Compared with Olaya et al. (2020), who benchmark multi-treatment uplift learners on ranking metrics, we evaluate the downstream budgeted *policy* with doubly robust off-policy evaluation and a positive control. Compared with Athey and Wager (2021), we compare their policy-learning approach empirically against estimate-then-optimise under a hard budget, rather than proposing a new learner.

Approach. We run the full chain on the Hillstrom e-mail experiment (Hillstrom, 2008), with an emphasis on credibility rather than on a new estimator. Every release freezes its configuration in a version-control commit *before* the code that uses it is run, in the spirit of Nosek, Ebersole, DeHaven, and Mellor (2018); real covariates and assignment are combined with simulated outcomes of known CATE as a positive control; and negative results are reported rather than tuned away.

Research question. *Can heterogeneous treatment effects estimated from a randomised experiment pick who gets which e-mail better than simple rules, at every budget, and if not, why not?* Section 2 formalises the problem, Section 3 presents the estimators and solvers, Section 4 the data and experimental design, Section 5 the main results, Section 6 the extensions and Section 7 the conclusions.

2 Problem Formulation

2.1 Causal Setting

Customer i has covariates X_i, receives arm $T_i \in \{0, 1, 2\}$ (no e-mail, Mens, Womens) and has potential outcomes $Y_i(a)$. Under SUTVA and randomisation with known propensity $e(a) = \mathbb{P}(T = a) = 1/3$, the conditional effect of arm a,

$$\tau_a(x) = \mathbb{E}\big[Y(a) - Y(0) \mid X = x\big], \tag{1}$$

is identified. The propensity is plugged in rather than estimated everywhere except in a deliberately confounded robustness check.

2.2 Budgeted Allocation

With $x_{ia} \in \{0, 1\}$ indicating that customer i receives arm $a \in \{1, 2\}$, costs c_a and budget B, the allocation problem is the multiple-choice knapsack (Sinha & Zoltners, 1979)

$$\begin{aligned} \max_{x} \quad & \textstyle\sum_i \sum_a \hat{\tau}_{ia}\, x_{ia} \\ \text{s.t.} \quad & \textstyle\sum_a x_{ia} \le 1 \;\; \forall i, \qquad \sum_i \sum_a c_a\, x_{ia} \le B. \end{aligned} \tag{2}$$

Hillstrom contains no cost data; we assume \$0.10 per Mens and \$0.08 per Womens e-mail and express a budget as a fraction f of the cost of e-mailing everyone at the average cost, $B = f \cdot n \cdot \bar{c}$ with $\bar{c} = \$0.09$.

2.3 Profit Objective

With gross margin m (assumed 30%; sensitivity 15% and 50%) and policy value $V(\pi)$, the incremental profit of a policy that assigns arm probabilities $\pi(a \mid x)$ is

$$\Pi(\pi) = m\,\big[V(\pi) - V(\pi_0)\big] - \tfrac{1}{n} \textstyle\sum_i \sum_a \pi(a \mid X_i)\, c_a, \tag{3}$$

where π_0 sends nothing. The *profit-targeted* policy replaces $\hat{\tau}_{ia}$ by $m\hat{\tau}_{ia} - c_a$ in (2), so it e-mails only customers whose expected margin exceeds the e-mail cost and may leave budget unspent.

2.4 Random Uptake and Chance Constraints

Suppose each e-mail carries a \$5 voucher, paid on every purchase by a targeted customer. With purchase probability p_{ia}, e-mailing i with arm a costs $c_a + 5p_{ia}$ in expectation with variance $25\, p_{ia}(1 - p_{ia})$, so the campaign cost $C(\pi)$ is random. A mean-budget policy only requires $\mathbb{E}[C] \le B$; a chance-constrained policy (Charnes & Cooper, 1959) requires

$$\mathbb{P}\big(C(\pi) > B\big) \le \varepsilon \iff \mathbb{E}[C(\pi)] + z_{1-\varepsilon}\sqrt{\mathrm{Var}[C(\pi)]} \le B, \tag{4}$$

with $\varepsilon = 0.05$. The implication holds only approximately: it uses a normal approximation to the sum of independent per-customer costs, accurate when thousands of customers are targeted.

2.5 Sequential Allocation

In the sequential variant, a campaign of 64,000 customers arrives in 16 batches of 4,000; the outcomes of a batch are observed before the next is allocated, and one total budget ($f = 0.3$) is paced evenly. The question becomes whether a policy should keep learning while it spends.

3 Theoretical Framework

3.1 Average Effects

For arm a against control we use the difference in means with a Welch interval and the fully interacted regression adjustment of Lin (2013),

$$Y_i = \alpha + \beta D_i + (X_i - \bar{X})^\top \gamma + D_i (X_i - \bar{X})^\top \delta + \varepsilon_i, \tag{5}$$

where $D_i = \mathbf{1}[T_i = a]$ and $\hat{\beta}$ estimates the ATE with HC2 standard errors; with past spend as the only covariate this is CUPED (Deng, Xu, Kohavi, & Walker, 2013).

3.2 CATE Learners

All base models are LightGBM regressors (Ke et al., 2017).

3.2.1 S- and T-learner

The S-learner fits $\mu(x, a)$ on $[x, \mathrm{onehot}(a)]$ and sets $\hat{\tau}_a(x) = \hat{\mu}(x, a) - \hat{\mu}(x, 0)$; the T-learner fits one outcome model $\hat{\mu}_a$ per arm.

3.2.2 X-learner

The X-learner (Künzel et al., 2019) imputes individual effects

$$\tilde{D}_i^1 = Y_i - \hat{\mu}_0(X_i) \;(T_i = a), \qquad \tilde{D}_i^0 = \hat{\mu}_a(X_i) - Y_i \;(T_i = 0), \tag{6}$$

regresses each on X and combines them as $\hat{\tau}_a = g\,\hat{\tau}_a^0 + (1 - g)\,\hat{\tau}_a^1$ with the known within-pair propensity $g = 1/2$.

3.2.3 DR-learner and Causal Forest

The DR-learner (Kennedy, 2023) regresses the difference $\Gamma_{ia} - \Gamma_{i0}$ of the doubly robust pseudo-outcomes

$$\Gamma_{ia} = \hat{\mu}_a(X_i) + \frac{\mathbf{1}[T_i = a]}{e(a)}\big(Y_i - \hat{\mu}_a(X_i)\big) \tag{7}$$

on X; the causal forest uses CausalForestDML from EconML (Athey et al., 2019; Battocchi et al., 2019). Both use the known propensity.

3.3 Ranking Metric

For arm a against control, sort customers by $\hat{\tau}_a$ and let $N_T(k), N_C(k)$ and $Y_T(k), Y_C(k)$ be the counts and outcome sums of treated and control customers among the top k. The Qini curve (Radcliffe, 2007) is

$$Q(k) = Y_T(k) - Y_C(k)\,\frac{N_T(k)}{N_C(k)}, \tag{8}$$

and the Qini coefficient is the area between the Qini curve and the random-targeting line, with the fraction targeted on $[0, 1]$ and the curve divided by the number of customers in the arm-vs-control test subsample, so it is in outcome units per customer ($ for spend, visit probability for visit).

3.4 Solving the Allocation Problem

Problem (2) is solved exactly with SCIP (Bestuzheva et al., 2021) through OR-Tools (Perron & Furnon, 2026). The Lagrangian heuristic (Everett, 1963) assigns $a_i^\star(\lambda) = \arg\max_{a\in\{0,1,2\}}\{\hat\tau_{ia} - \lambda c_a\}$ (value 0 for no e-mail) and bisects $\lambda \geq 0$ until the spend fits B; the dual function

$$L(\lambda) = \textstyle\sum_i \max\big(0, \max_a(\hat\tau_{ia} - \lambda c_a)\big) + \lambda B \qquad (9)$$

upper-bounds the optimum. Leftover budget is spent greedily by $\hat\tau_{ia}/c_a$ ("repair"), which fixes the all-or-nothing behaviour of the heuristic on groups with tied estimates.

3.5 Off-Policy Evaluation

With known propensity, the inverse-propensity (Horvitz & Thompson, 1952) and doubly robust (Dudík et al., 2011; Robins et al., 1994) estimates of a policy's value are

$$\hat V_{\mathrm{IPW}}(\pi) = \frac{1}{n}\sum_i \frac{\pi(T_i \mid X_i)}{e(T_i)}\, Y_i, \qquad (10)$$

$$\hat V_{\mathrm{DR}}(\pi) = \frac{1}{n}\sum_i \Big[\sum_a \pi(a \mid X_i)\,\hat\mu_a(X_i) + \frac{\pi(T_i \mid X_i)}{e(T_i)}\big(Y_i - \hat\mu_{T_i}(X_i)\big)\Big]. \qquad (11)$$

Both are unbiased for any fixed $\hat\mu$ because e is known.

3.5.1 Repeated Cross-Fitting

To use all customers and include model-refit variance, the policy is also evaluated by K-fold cross-fitting (Chernozhukov et al., 2018): models are fitted on $K-1$ folds and the policy is allocated and scored on the held-out fold. Repeating with R fold splits gives $\hat\theta_r$ and $\hat\sigma_r$, combined as

$$\hat\theta = \mathrm{median}_r\,\hat\theta_r, \qquad \hat\sigma^2 = \mathrm{median}_r\,\big(\hat\sigma_r^2 + (\hat\theta_r - \hat\theta)^2\big). \quad (12)$$

3.6 Direct Policy Learning

Doubly robust policy learning (Athey & Wager, 2021) skips the CATE model and maximises

$$\hat\pi = \arg\max_{\pi\in\Pi} \sum_i \Big[m\,\Gamma_{i,\pi(X_i)} - (1+\lambda)\,c_{\pi(X_i)}\Big] \qquad (13)$$

over depth-2 and depth-3 policy trees (EconML `PolicyTree`), with Γ from (7) cross-fitted inside each training fold and λ the smallest grid value whose training spend fits the budget.

3.7 Correcting Selection Bias in Uptake

Two corrections of the uptake estimate $\hat p$ used by the chance constraint are considered. *Recalibration* lets the allocator select customers on a held-out half of the training data and estimates

$$\hat\kappa = \max\Big(1,\ \frac{\sum_{i\in\mathcal{S}} \mathbf{1}[T_i = a_i]\,\mathrm{Conv}_i/e(a_i)}{\sum_{i\in\mathcal{S}} \hat p_{ia_i}}\Big), \qquad (14)$$

the ratio of IPW-estimated to predicted purchases among the selected set $\mathcal{S}$, then plans with $\hat\kappa\,\hat p$. The *bootstrap bound* refits $\hat p$ on 20 bootstrap resamples and plans with the per-customer 90% quantile.

3.8 Thompson Sampling

The bandit refits the S-learner after every batch on a bootstrap resample of all logged data (Eckles & Kaptein, 2014; Thompson, 1933) and allocates on the resulting $\hat\tau$ with 5% uniform exploration. It is compared with experiment-then-target, which runs a uniform RCT on the first $k \in \{1, 2, 4\}$ batches and then fixes its model.

4 Methodology

4.1 Data

The MineThatData challenge data (Hillstrom, 2008) contain 64,000 customers who purchased in the previous twelve months, randomised in equal thirds to a men's-merchandise e-mail ("Mens"), a women's-merchandise e-mail ("Womens") or no e-mail. Covariates are recency, past-year spend, indicators for past men's and women's purchases, a new-customer flag, zip-code type and purchase channel. Outcomes over two weeks are a website visit (base rate 14.7%), a conversion (0.9%) and spend (mean $1.05; mean $116 among buyers). The data were obtained through `scikit-uplift` (scikit-uplift developers, 2022) and verified against a recorded SHA-256 hash. Balance is excellent: the largest standardised mean difference between a treated arm and control across 11 encoded covariates is 0.014. A single 70/30 split, stratified by arm, gives 44,800 training and 19,200 test customers; everything that is fitted or selected uses training data only.

4.2 Pre-registration Protocol

The study was carried out as six releases (v1.0 to v1.5). For each, the configuration (seeds, splits, grids, costs, budgets) was committed before any code that uses it was written, and changes made after seeing results are logged with a reason.

4.3 Evaluation Metrics

4.3.1 Ranking quality

Qini coefficients on the test split, with percentile bootstrap intervals (Efron & Tibshirani, 1993) and a permutation null of 500 random orderings as the random baseline.

4.3.2 Policy value

Incremental spend, visit or profit per customer against sending nothing, estimated with (11); random targeting and "best single arm" baselines are evaluated as stochastic policies whose expected cost equals the budget.

4.3.3 Recovery of the truth

On simulated outcomes: the correlation of $\hat{\tau}$ with the true τ, the known true value of each policy, and the true probability of a budget overrun.

4.4 Positive Control

The positive control keeps the real X_i and T_i and simulates outcomes. Baseline visit and purchase probabilities rise with standardised log past spend z_i, $v_0(x) = 0.10\, e^{0.3z}$ and $p_0(x) = 0.006\, e^{0.3z}$; spend is a purchase indicator times a LogNormal amount with mean \$100 ($\sigma = 1$), so $\tau_a^{\text{spend}}(x) = 100 \cdot \delta_a^p(x)$. The Mens e-mail adds 0.2 percentage points (pp) of purchase probability everywhere; the Womens e-mail adds 3 pp, but only in the segment $S = \{\texttt{womens} = 1, \texttt{recency} \le 6\}$ (32% of customers). The simulated data match the real base rates closely (visit 13.3%, purchase 1.0%, mean spend \$1.04).

4.5 Experimental Design

4.5.1 Heterogeneity sweep

The Womens effect keeps its customer average $\bar{\delta}$ while a heterogeneity parameter $h \in [0, 1]$ concentrates it in S (share $s = 0.32$):

$$\delta_{\text{in}} = \bar{\delta}\Big(1 + h\,\frac{1-s}{s}\Big), \qquad \delta_{\text{out}} = \bar{\delta}\,(1-h). \tag{15}$$

Each of $5 \times 4 \times 20$ replications ($h \in \{0, 0.25, 0.5, 0.75, 1\}$, $n \in \{4, 16, 64, 256\}$ thousand) resamples customers, re-randomises arms, fits the S-learner, allocates at a 30% budget and records the true gain and the analyst's DR estimate with an influence-function interval.

4.5.2 Cross-fitted evaluations

Profit and policy learning use 5 folds with 10 and 5 repeats; the chance-constraint studies use 15 cross-fitted campaigns of about 12,800 customers, with overruns counted by Monte Carlo against the true uptake.

4.5.3 Sequential simulation

20 replications per environment (heterogeneous $h = 1$ and homogeneous $h = 0$), with coupled potential outcomes so that all policies face identical customers and outcome draws.

Table 1: Average effects vs. no e-mail on all 64,000 customers (difference in means, 95% CI; visit and conversion in percentage points, pp).

Outcome	Mens e-mail	Womens e-mail
Spend (\$)	0.77 [0.49, 1.05]	0.42 [0.17, 0.68]
Visit (pp)	7.7 [7.0, 8.3]	4.5 [3.9, 5.2]
Conversion (pp)	0.68 [0.50, 0.86]	0.31 [0.15, 0.47]

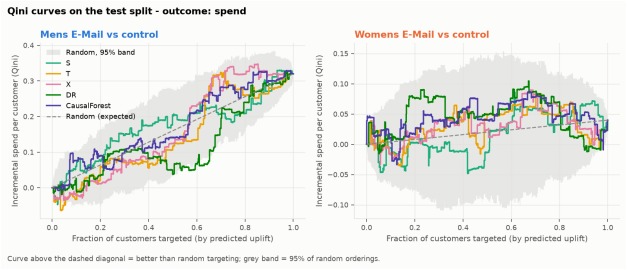


Figure 1: Qini curves for spend on the test split. Every learner stays inside the 95% band of random orderings (grey).

4.6 Statistical Reporting and Reproducibility

Every number carries a 95% interval (bootstrap, analytic or across replications). Code, frozen configurations, a decision log and all results are public at `https://github.com/GrafZemeliSeverochecanskih/budget-uplift`; a single command reproduces every table and figure, a fresh-clone run reproduced every result table bit-for-bit (largest numeric difference 2×10^{-16}, wall-clock timings excluded), and 125 unit tests check every estimator against simulated data with a known effect.

5 Results

5.1 Average Effects

Both e-mails raise every outcome (Table 1). Regression adjustment narrows the spend interval by at most 0.12% (visit: 1.5%), because pre-period covariates explain almost none of two-week spend. Detecting the observed spend effect at 80% power needs 5,953 customers per arm for Mens but 15,808 for Womens (range 6,160 to 99,739 over the effect's interval); the minimum detectable spend effect with the available ≈21,300 per arm is \$0.37 to \$0.41.

5.2 Targeting on Real Data

5.2.1 No learnable signal for spend

All ten learner×arm Qini intervals for spend contain zero and every one-sided permutation p-value is at least 0.24 (Table 2, Figure 1). On *visit*, all five learners rank the Womens effect better than random ($p = 0.004$), whereas the Mens effect is not targetable.

5.2.2 Targeted policies do not beat simple rules

The targeted policy is indistinguishable from random targeting at every budget and its point estimate lies *below* "Mens

Table 2: Qini coefficients on the test split (95% bootstrap CI). Random: mean and 95% band of 500 random orderings.

(a) Spend ($)

Learner	Mens	Womens
S	0.013 [−0.078, 0.103]	−0.005 [−0.079, 0.067]
T	−0.016 [−0.096, 0.057]	0.010 [−0.068, 0.093]
X	0.001 [−0.078, 0.080]	0.010 [−0.065, 0.088]
DR	−0.038 [−0.127, 0.042]	0.026 [−0.056, 0.112]
C. forest	0.012 [−0.058, 0.100]	0.024 [−0.054, 0.109]
Random	−0.005 [−0.086, 0.073]	0.000 [−0.062, 0.065]

(b) Visit

Learner	Mens	Womens
S	0.0004 [−0.0014, 0.0021]	0.0032 [0.0016, 0.0050]
T	0.0005 [−0.0014, 0.0021]	0.0027 [0.0010, 0.0044]
X	0.0001 [−0.0017, 0.0019]	0.0027 [0.0011, 0.0043]
DR	0.0000 [−0.0017, 0.0016]	0.0026 [0.0008, 0.0043]
C. forest	0.0006 [−0.0011, 0.0023]	0.0030 [0.0014, 0.0048]
Random	−0.0001 [−0.0019, 0.0019]	0.0000 [−0.0016, 0.0018]

e-mail to a random subset" throughout (Figure 2, left). At $f = 1$ it earns +$0.31 [−0.18, 0.83] per customer against +$0.56 [0.11, 1.03] for the Mens rule; the paired difference is −$0.25 [−0.59, 0.09]. On visit the targeted policy beats random targeting at 7 of 10 budgets but the Mens rule at only one.

5.2.3 Solvers

SCIP proves optimality of (2) on all 33 instances up to $n =$ 64,000 (median 49.3 s at the largest size). The Lagrangian heuristic is always feasible, within 0.34% of the optimum and about 170× faster (0.29 s). Both are single-run wall-clock times on one machine (AMD Ryzen 7 7435HS, 8 cores, 16 GB RAM) and vary between runs. CP-SAT finds no feasible solution within 30 s for 9 of 10 budgets at $n =$ 10,000; a plausible explanation is that the multiple-choice knapsack has a near-integral LP relaxation, which LP-based branch-and-bound exploits and CP-SAT does not; we did not test this directly.

5.3 Positive Control

The unchanged pipeline finds the planted effect (Table 3, Figure 2, right). For the Womens arm all five learners beat the permutation null ($p \leq 0.022$) with Qini values indistinguishable from an oracle that ranks by the true τ, and estimated effects correlate with the truth at 0.62 to 0.83 (spend) and 0.90 to 0.96 (visit). For the Mens arm the true effect is constant, so every learner *and* the oracle have Qini near zero. The targeted policy beats random targeting at 8 of 10 budgets for spend (+$0.50 [0.27, 0.75] at $f = 0.3$) and at all budgets for visit, and its true value is 90 to 98% of the oracle's. DR intervals cover the true policy value in 73 of 80 cases. At a nominal 95%, 76 of 80 would be expected; under independence, 73 or fewer would occur with probability 0.105, so 73/80 is not evidence of under-coverage. The 80 intervals are also strongly dependent: all seven misses concern near-identical visit policies on the same test draw, which lies about two standard errors high.

Table 3: Positive control, Womens arm: Qini (spend, $) and correlation of $\hat{\tau}$ with the true τ.

Learner	Qini, spend	ρ, spend	ρ, visit
S	0.076 [−0.027, 0.164]	0.83	0.96
T	0.095 [0.019, 0.166]	0.65	0.92
X	0.100 [0.029, 0.175]	0.75	0.94
DR	0.111 [0.042, 0.178]	0.62	0.91
C. forest	0.089 [0.011, 0.166]	0.65	0.90
Oracle	0.098 [0.013, 0.171]	1.00	1.00

Table 4: Visit effect (pp) in the deliberately confounded sample: RCT benchmark, naive difference in means, IPW with an estimated propensity and AIPW.

Arm	RCT	Naive	IPW	AIPW
Mens	7.66	9.44	7.80	7.78
Womens	4.52	5.91	4.75	4.70

5.4 When Does Targeting Pay Off?

Against the best single arm, which isolates the value of heterogeneity, targeting truly pays from $h = 0.25$ when $n \geq 16{,}000$ (e.g. +$0.06 [0.05, 0.08] per customer at $n =$ 64,000) and at every n from $h = 0.5$, while at $h = 0$ it earns nothing (−$0.007 [−0.013, 0.000]; Figure 3). Detection lags far behind: at Hillstrom's size a single experiment's DR interval excludes zero in only 5%, 25%, 85% and 95% of replications for $h = 0.25, 0.5, 0.75, 1$. Between "targeting would earn more" and "one can show that it does" lies a band in which the policy is worth $0.04 to $0.23 per customer yet looks like noise. The Hillstrom null is therefore weak evidence *against* heterogeneity, all the more because Hillstrom's real Womens effect ($0.42) is less than half the simulated average ($0.96).

5.5 Robustness Checks

A deliberately confounded sample biases the naive visit effect by +1.8 pp; IPW with an estimated propensity and AIPW recover the RCT value (Table 4) within 0.25 pp, AIPW within 0.2 pp (largest deviations 0.22 pp for IPW and 0.18 pp for AIPW). Allocating on a lower confidence bound of $\hat{\tau}$ lowers both mean and 5th-percentile policy value at every budget. Sentence-embedding features (Reimers & Gurevych, 2019) change the spend Qini by +0.019 [−0.042, 0.081] for the Mens arm and −0.022 [−0.052, 0.010] for the Womens arm. No gain is expected: each sentence is generated from the same tabular covariates, so the embedding adds no new information, only a different representation.

6 Extensions

The pre-registered extensions below reuse the pipeline of Section 5.

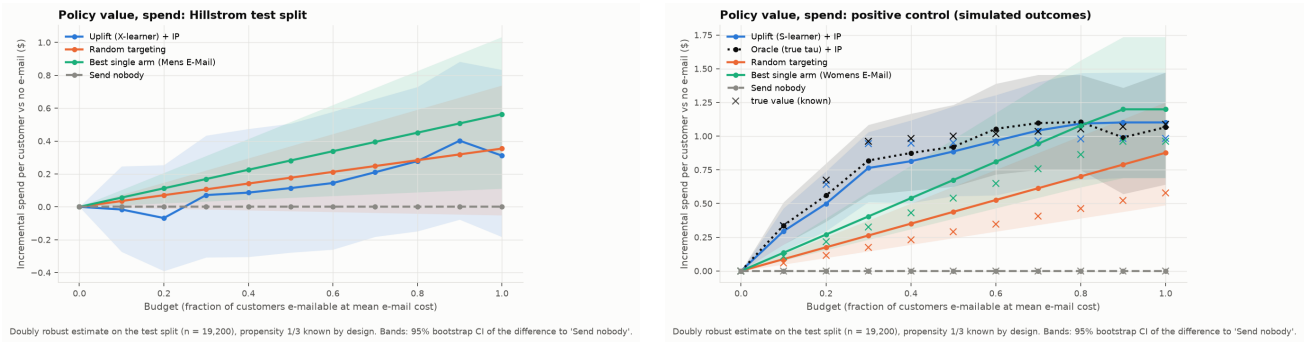


Figure 2: Incremental spend per customer vs. budget (doubly robust, 95% bootstrap CI). Left: real Hillstrom outcomes. Right: positive control with the same covariates, assignment and code; × marks the known true value of each policy.

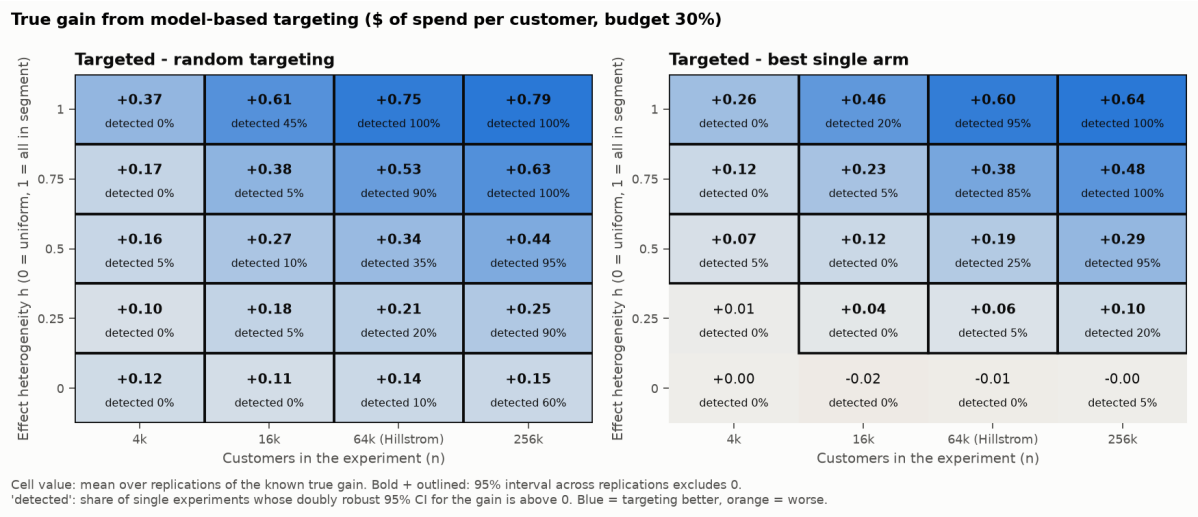


Figure 3: True gain from targeting over heterogeneity h and sample size n (mean of 20 replications). Outlined cells: 95% interval excludes zero. "Detected": share of single experiments whose DR interval lies above zero.

6.1 Profit and Cross-Fitting

With a profit objective (Figure 4) the targeted policy chooses its own budget: on the positive control its profit-optimal budget is $f = 0.4$ [0.4, 0.65], matching the true optimum, and when unconstrained it stops at \$0.047 of spend per customer while earning \$0.211 [0.132, 0.290] (true \$0.240), more than revenue targeting (\$0.173, true \$0.217). On real Hillstrom no policy separates from any other (0 of 33 paired comparisons) and the best point estimate is the simplest rule, Mens e-mail to everyone (\$0.130 [0.045, 0.215]). Repeated cross-fitting (12) narrows intervals by 37 to 58% relative to the test split while including refit variance, covers the true profit in 142 of 143 positive-control cases, and establishes that e-mailing pays on real data (+\$0.65 [0.37, 0.93] revenue per customer at $f = 1$), but not that *targeting* pays.

6.2 Policy Learning

Doubly robust policy trees never beat estimate-then-optimise (Figure 5). On real data all paired differences contain zero, and the depth-2 tree fitted to all customers reads "Mens e-mail to everyone except low-spend, very recent customers". On the positive control, trees lose clearly at a tight budget (−\$0.12 [−0.18, −0.07] at $f = 0.2$): the profitable segment holds 32% of customers and a tree can only e-mail whole leaves.

6.3 Chance Constraints

6.3.1 The optimiser's curse

Under random voucher uptake, the mean-budget policy overruns its budget in 63 to 85% of campaigns and the plug-in chance-constrained policy in 14 to 32% (Figure 6), at an 8 to 18% loss of profit, missing the 5% target it believes it meets. Estimated uptake is calibrated overall (0.012 vs. 0.012) but about 10% too low among the customers the allocator selects (0.026 vs. 0.028 to 0.029): customers who look cheap are partly cheap by estimation error (Smith & Winkler, 2006).

6.3.2 Repairing the curse

Recalibration barely helps (Table 5, Figure 6): $\hat{\kappa}$ comes out at about 1 while the true ratio among selected customers is 1.08 to 1.12, because a 10% bias is below what IPW on a few hundred matched purchases can resolve. The bootstrap bound holds the target with room to spare (overrun at most 0.8%) but gives up 16 to 19% of profit.

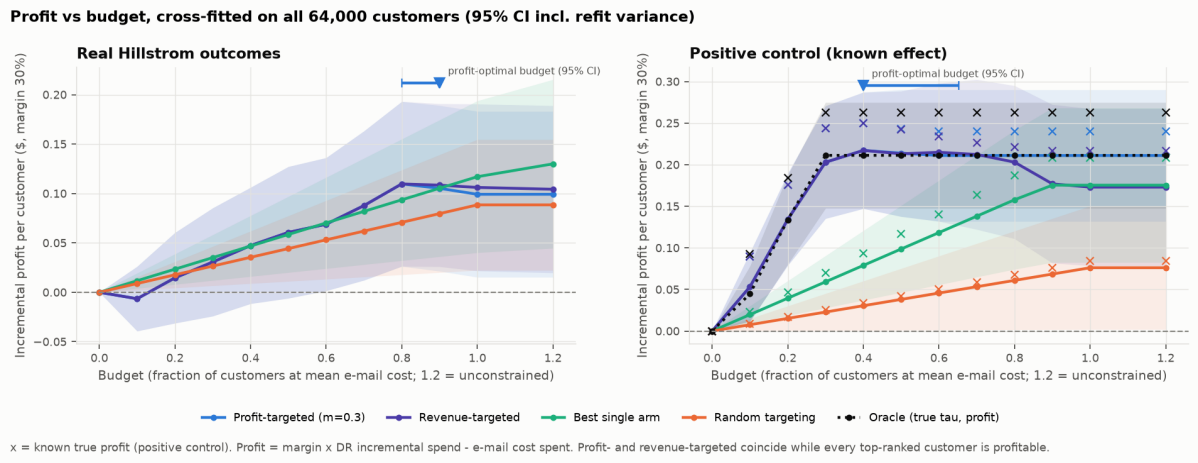


Figure 4: Incremental profit vs. budget at a 30% margin, cross-fitted on all customers (5 folds × 10 repeats). Triangle: profit-optimal budget with 95% bootstrap interval.

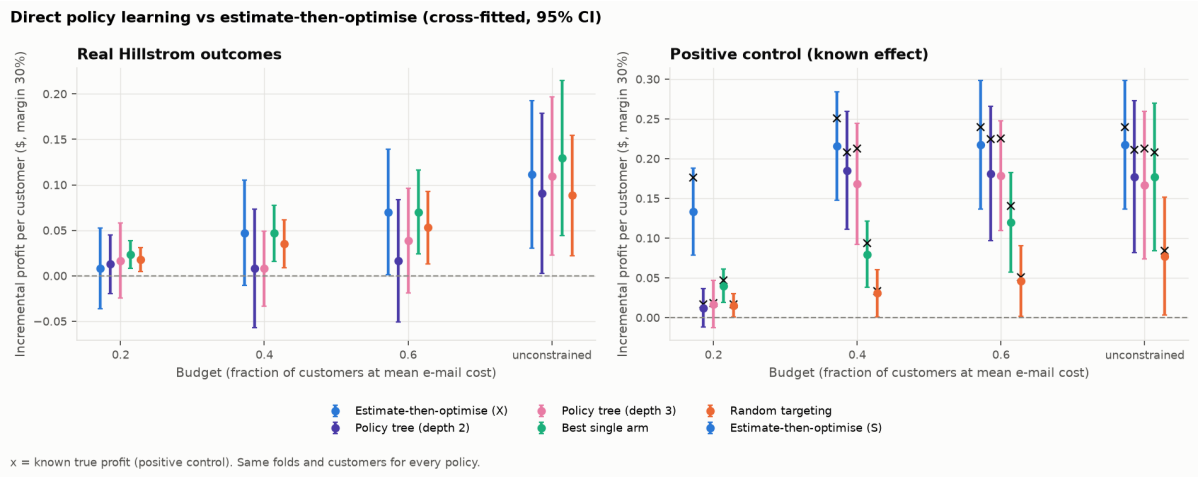


Figure 5: Policy trees vs. estimate-then-optimise on identical cross-fitting folds (5×5).

6.4 Sequential Allocation

The bandit earns 68% of the oracle's profit when effects are heterogeneous, against 47 to 60% for experiment-then-target (Table 6, Figure 7). It beats $k = 1$ by \$0.027 [0.001, 0.054] and $k = 4$ by \$0.054 [0.046, 0.062] per customer; against $k = 2$ the gain, \$0.021 [−0.002, 0.043], is not significant. Its main advantage is robustness: an interval four times narrower than $k = 1$'s. With homogeneous effects the bandit is indistinguishable from $k \in \{1, 2\}$, so adaptivity costs nothing when there is nothing to learn.

7 Conclusions

7.1 Summary of Contributions

We deliver an end-to-end, multi-arm, budget-constrained uplift pipeline with doubly robust evaluation, run under a pre-registration-by-commit protocol. Its central finding on Hillstrom is negative for spend; a positive control shows that this is a property of the data rather than of the method; and a heterogeneity sweep quantifies how weak that negative evidence is. Five further studies show that a profit objective chooses its own budget, that repeated cross-fitting sharpens policy evaluation, that policy trees do not beat estimate-then-optimise, that plug-in chance constraints fail through the optimiser's curse while a conservative bound holds, and that a budgeted bandit beats experiment-then-target with $k = 1$ and $k = 4$ and is not significantly different from $k = 2$.

7.2 Discussion

For practitioners the results suggest four habits: evaluate targeted policies against the best single arm, not only against random targeting; size experiments for *policy* differences, which are much harder to detect than average effects; distrust plug-in constraints built on the same model that selected the customers; and prefer a conservative bound to a data-driven correction when uptake is rare.

7.3 Limitations

SUTVA is assumed throughout (no household spill-over or cannibalisation). Outcomes cover two weeks after a single 2008 campaign, with no long-run effects, fatigue or unsubscribes. Costs, margin and voucher value are assumptions.

Table 5: True overrun probability on the positive control (95% CI, 15 campaigns) for the plug-in, recalibrated and bootstrap-bound chance constraints, and true profit for the plug-in constraint vs. the bootstrap bound (true expected profit, $ per customer).

	Overrun probability			Profit	
Budget	Plug-in	Recalibrated	Bootstrap	Plug-in	Bootstrap
0.02	0.14 [0.08, 0.20]	0.11 [0.05, 0.17]	0.008	0.032	0.026
0.04	0.22 [0.14, 0.30]	0.19 [0.11, 0.27]	0.003	0.071	0.058
0.06	0.29 [0.21, 0.36]	0.21 [0.12, 0.30]	0.001	0.112	0.092
0.08	0.32 [0.25, 0.40]	0.24 [0.15, 0.34]	0.000	0.150	0.126

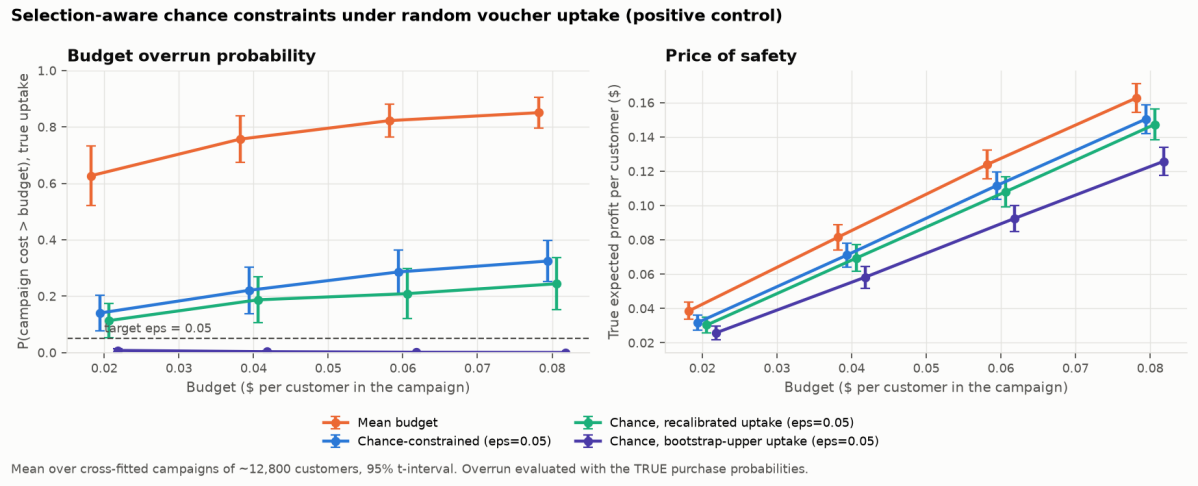


Figure 6: Overrun probability (true uptake) and true profit on the positive control, 15 campaigns: mean budget and the plug-in, recalibrated and bootstrap-bound chance constraints.

Table 6: Total true profit per customer over a 64,000-customer campaign (95% CI, 20 replications). The interval for $k = 4$ (homogeneous) is [0.0448, 0.0453] at four decimals: after the RCT on four batches the replications target almost identically (785 to 814 e-mails per batch), so their profit barely varies.

Policy	Heterogeneous	Homogeneous
Oracle (true τ)	0.263	0.070
Thompson bandit	**0.178** [0.172, 0.184]	0.054 [0.050, 0.058]
Exp.-then-target, $k = 1$	0.150 [0.123, 0.177]	0.052 [0.041, 0.064]
Exp.-then-target, $k = 2$	0.157 [0.135, 0.179]	0.050 [0.044, 0.056]
Exp.-then-target, $k = 4$	0.124 [0.117, 0.131]	0.045 [0.045, 0.045]
Random targeting	0.025	0.025

The positive control is only as realistic as its generator: additive effects, a simple segment rule on observed covariates, and amounts independent of X. Learners and LightGBM parameters were fixed from the first release onwards. Several comparisons across budgets and outcomes are reported without multiplicity correction.

7.4 Future Work

A scale check on the Criteo uplift data (Diemert, Betlei, Renaudin, & Amini, 2018) was planned but not run because all known download locations were unavailable. Other natural extensions are adaptive budget pacing for the bandit, uptake models calibrated on a dedicated holdout, a profit objective with returns and long-run effects, and multiplicity-adjusted comparisons across budgets.

8 Broader Impact Statement

Targeting decides who receives an offer, so it can also decide who does not: a profit-optimal policy may systematically exclude groups that look less responsive, and covariates can act as proxies for protected characteristics. We use only purchase-history covariates from a public, anonymised dataset, but any deployment should audit who is excluded and why. The main finding cuts the other way as well: on these data, simple rules perform as well as targeted ones, so the extra data collection and modelling that targeting requires would not be justified. More generally, the preregistration protocol and the public decision log are offered as a way to make analytics claims checkable rather than persuasive.

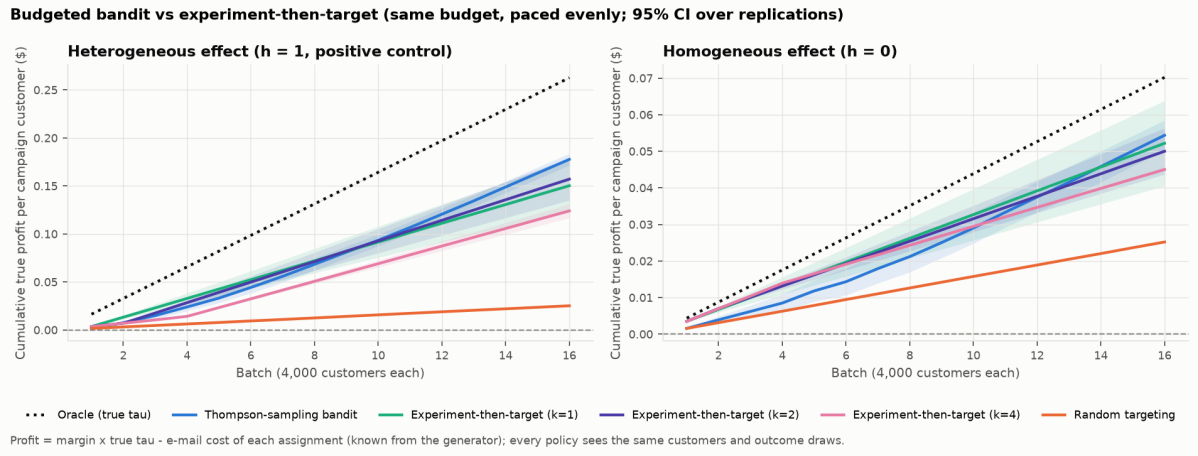


Figure 7: Cumulative true profit per campaign customer over 16 batches (95% CI over replications).

References

Athey, S., Tibshirani, J., & Wager, S. (2019). Generalized random forests. *The Annals of Statistics*, *47*(2), 1148-1178.

Athey, S., & Wager, S. (2021). Policy learning with observational data. *Econometrica*, *89*(1), 133-161.

Badanidiyuru, A., Kleinberg, R., & Slivkins, A. (2018). Bandits with knapsacks. *Journal of the ACM*, *65*(3), 13:1-13:55.

Battocchi, K., Dillon, E., Hei, M., Lewis, G., Oka, P., Oprescu, M., & Syrgkanis, V. (2019). *EconML: A Python package for ML-based heterogeneous treatment effects estimation.* Retrieved from `https://github.com/py-why/EconML` (Version 0.17.0 used)

Bestuzheva, K., et al. (2021). *The SCIP Optimization Suite 8.0.* arXiv:2112.08872.

Charnes, A., & Cooper, W. W. (1959). Chance-constrained programming. *Management Science*, *6*(1), 73-79.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, *21*(1), C1-C68.

Deng, A., Xu, Y., Kohavi, R., & Walker, T. (2013). Improving the sensitivity of online controlled experiments by utilizing pre-experiment data. In *Proceedings of the sixth acm international conference on web search and data mining (wsdm)* (p. 123-132).

Diemert, E., Betlei, A., Renaudin, C., & Amini, M.-R. (2018). A large scale benchmark for uplift modeling. In *Proceedings of the adkdd and targetad workshop, kdd.*

Dudík, M., Langford, J., & Li, L. (2011). Doubly robust policy evaluation and learning. In *Proceedings of the 28th international conference on machine learning (icml)* (p. 1097-1104).

Eckles, D., & Kaptein, M. (2014). *Thompson sampling with the online bootstrap.* arXiv:1410.4009.

Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall/CRC.

Everett, H. (1963). Generalized Lagrange multiplier method for solving problems of optimum allocation of resources. *Operations Research*, *11*(3), 399-417.

Gutierrez, P., & Gérardy, J.-Y. (2017). Causal inference and uplift modelling: A review of the literature. In *Proceedings of the 3rd international conference on predictive applications and apis* (Vol. 67, p. 1-13).

Hillstrom, K. (2008). *The MineThatData e-mail analytics and data mining challenge.* MineThatData blog. Retrieved from `https://blog.minethatdata.com/2008/03/minethatdata-e-mail-analytics-and-data.html`

Horvitz, D. G., & Thompson, D. J. (1952). A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, *47*(260), 663-685.

Imbens, G. W., & Rubin, D. B. (2015). *Causal inference for statistics, social, and biomedical sciences: An introduction.* Cambridge University Press.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., … Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in neural information processing systems 30 (neurips)* (p. 3146-3154).

Kellerer, H., Pferschy, U., & Pisinger, D. (2004). *Knapsack problems*. Springer.

Kennedy, E. H. (2023). Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, *17*(2), 3008-3049.

Kitagawa, T., & Tetenov, A. (2018). Who should be treated? Empirical welfare maximization methods for treatment choice. *Econometrica*, *86*(2), 591-616.

Künzel, S. R., Sekhon, J. S., Bickel, P. J., & Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, *116*(10), 4156-4165.

Lin, W. (2013). Agnostic notes on regression adjustments to experimental data: Reexamining Freedman's critique. *The Annals of Applied Statistics*, *7*(1), 295-318.

Nie, X., & Wager, S. (2021). Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, *108*(2),

299-319.

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, *115*(11), 2600-2606.

Olaya, D., Coussement, K., & Verbeke, W. (2020). A survey and benchmarking study of multitreatment uplift modeling. *Data Mining and Knowledge Discovery*, *34*, 273-308.

Perron, L., & Furnon, V. (2026). *OR-Tools.* Version 9.15.6755. Retrieved from `https://developers.google.com/optimization`

Radcliffe, N. J. (2007). Using control groups to target on predicted lift: Building and assessing uplift models. *Direct Marketing Analytics Journal*, *1*, 14-21.

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 conference on empirical methods in natural language processing (emnlp-ijcnlp)* (p. 3982-3992).

Robins, J. M., Rotnitzky, A., & Zhao, L. P. (1994). Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, *89*(427), 846-866.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, *66*(5), 688-701.

Russo, D. J., Van Roy, B., Kazerouni, A., Osband, I., & Wen, Z. (2018). A tutorial on Thompson sampling. *Foundations and Trends in Machine Learning*, *11*(1), 1-96.

scikit-uplift developers. (2022). *scikit-uplift: uplift modeling in scikit-learn style in Python.* Version 0.5.1. Retrieved from `https://github.com/maks-sh/scikit-uplift`

Sinha, P., & Zoltners, A. A. (1979). The multiple-choice knapsack problem. *Operations Research*, *27*(3), 503-515.

Smith, J. E., & Winkler, R. L. (2006). The optimizer's curse: Skepticism and postdecision surprise in decision analysis. *Management Science*, *52*(3), 311-322.

Thompson, W. R. (1933). On the likelihood that one unknown probability exceeds another in view of the evidence of two samples. *Biometrika*, *25*(3/4), 285-294.

Wager, S., & Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, *113*(523), 1228-1242.

Zhao, Y., Fang, X., & Simchi-Levi, D. (2017). Uplift modeling with multiple treatments and general response types. In *Proceedings of the 2017 siam international conference on data mining* (p. 588-596).

A Settings

Table 7 lists the settings of every study, taken from the frozen configuration file `config/config.yaml`.

Table 7: Hyperparameters, seeds and grids (from `config/config.yaml`).

Setting	Value
Seed and split	Seed 20261006; 70/30 train/test split, stratified by arm
Costs and budgets	\$0.10 per Mens and \$0.08 per Womens e-mail; budget fractions 0 to 1 in steps of 0.1 (profit: also 1.2)
Inference	500 bootstrap replications; $\alpha = 0.05$
LightGBM	Grid `num_leaves` $\{7, 31\}$, `min_child_samples` $\{100, 400\}$, `n_estimators` $\{100, 300\}$, 3-fold CV on training data; fixed `learning_rate` 0.03, `subsample` 0.8, `colsample_bytree` 0.8; chosen for spend: 7 / 400 / 100
Causal forest	400 trees, minimum leaf size 50
Profit	Margin 0.30 (sensitivity 0.15 and 0.50); cross-fitting 5 folds × 10 repeats
Policy learning	Cross-fitting 5 folds × 5 repeats; depths $\{2, 3\}$; minimum leaf size 200; $\lambda \in \{0\} \cup 24$ log-spaced values in $[0.01, 20]$
Chance constraints	Voucher \$5; $\varepsilon = 0.05$; budgets 0.02 to 0.08 per customer; 5 folds × 3 repeats (15 campaigns); 2,000 Monte Carlo uptake draws
Selection-aware correction	2 inner calibration folds; $\hat{\kappa}$ floored at 1; 20 bootstrap refits; 90% quantile
Heterogeneity sweep	$h \in \{0, 0.25, 0.5, 0.75, 1\}$; $n \in \{4{,}000, 16{,}000, 64{,}000, 256{,}000\}$; 20 replications; budget 0.3
Sequential	64,000 customers in batches of 4,000; budget 0.3; $k \in \{1, 2, 4\}$; bandit exploration $\varepsilon = 0.05$; 20 replications